

Viha Tripathi

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EDUCATION

Mcmaster University

Electrical Engineering

Hamilton, ON

September 2023 – May 2027

TECHNICAL SKILLS

Programming/tools: Python, C/C++, MATLAB, Verilog/VHDL, JavaScript, HTML/CSS, R, SQL, Git, Visual Studio, PyCharm, Microsoft Office (Word, Excel, PowerPoint), Google Workspace

Hardware/Design: Assembly, Embedded Systems, Power Systems, Circuit Design and Hardware Interfacing, PLC Programming, Electrical Schematics (ORCAD PSpice, LTSpice), Micro-controller Programming, Function Generators, Bread-boarding, Power Supply, Oscilloscopes, Digital Multimeter, 3D CAD (AutoCAD, Solidworks), 3D Printing, Analog Discovery 3, Laser printing

PROJECTS

Embedded Spatial Mapping System | C, C++, Python, AutoCAD

January 2025 – March 2025

- Designed and implemented an **embedded system** to map indoor environments using a **Time-of-Flight (ToF)** sensor and **stepper motor**, creating a cost-effective alternative to commercial **LiDAR** systems
- Developed **microcontroller firmware** to interface with the VL53L1X **ToF** sensor via **I2C**, enabling precise 360° spatial measurements and serial data transmission through **UART** for real-time processing
- Processed and visualized acquired spatial data using **Python**, generating 3-D reconstructions of scanned environments for real-world application in robotics and navigation
- Designed a custom sensor housing and mounting assembly in AutoCAD** to align the ToF sensor, stepper motor, and PCB, ensuring structural stability and accurate scanning
- Created **cable schedules** and a detailed **bill of materials** to support organized wiring, part tracking, and hardware integration

Voltage-Controlled Switch Design | LTSpice, Breadboarding, Oscilloscope

February 2025 – March 2025

- Designed and implemented two **voltage-controlled analog switches** under a +5V supply, creating **electrical schematics** in **LTSpice** and simulating their behaviour, then validating on a **breadboard**
- Switch 1 opened a connections based on control voltage, identifying and minimizing leakage current and ON-resistance, contributing to more reliable switching in real-world analog control circuits
- Switch 2 toggled voltage between two paths based on a digital signal, enabling dynamic signal selection

EXPERIENCE

Second-Year Representative

July 2024 - Present

Electrical and Computer Engineering Society

Hamilton, ON

- Represented **250+** second-year students in department meetings by gathering concerns about course structure and workload imbalances, prompting faculty to re-evaluate scheduling for core labs
- Identified a lack of transparency around academic support resources, then initiated and organized a department-wide Q&A event with professors and upper-year mentors, increasing student engagement by **40%**
- Collaborated with ECE faculty to propose changes to tutorial delivery based on peer feedback, leading to the implementation of hybrid tutorials in key second-year courses

Marketing Executive

June 2024 – Present

DeltaHacks

Hamilton, ON

- Judged **30+** hackathon presentations, providing constructive feedback to teams and contributing to the overall success of the event
- Engaged **600+** participants and improved communication efficiency through targeted social media content
- Collaborated with a team of **30+** to address participant inquiries, ensuring smooth operations
- Organized and promoted events, enhancing brand visibility through **strategic marketing** campaigns

Vice President of Design

May 2024 – Present

McMaster Society of Engineering Research

Hamilton, ON

- Led the creation of visual content for outreach and advertising campaigns using **Figma**, designing promotional materials to effectively market society events and initiatives
- Simplified and communicated the concept of engineering research, increasing awareness and engagement across diverse audiences at McMaster through clear messaging and visuals